

PMS ASSIGNMENT-3

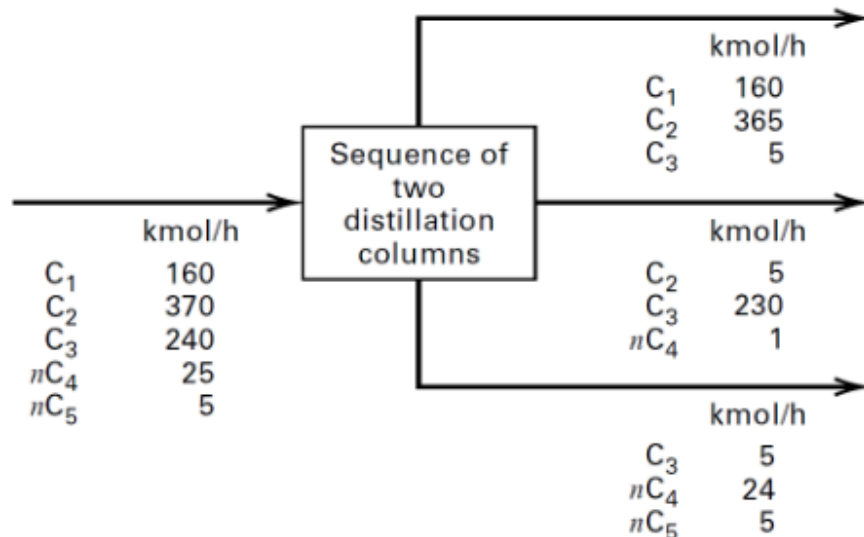
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ROLL NO. : 22CH3FP19

Q1: SEQUENCING OF DISTILLATION COLUMN

Two distillation columns are used to produce the products indicated in the Figure. Establish the type of condenser and an operating pressure for each column for the:

(a) direct sequence (C₂/C₃ separation first) and

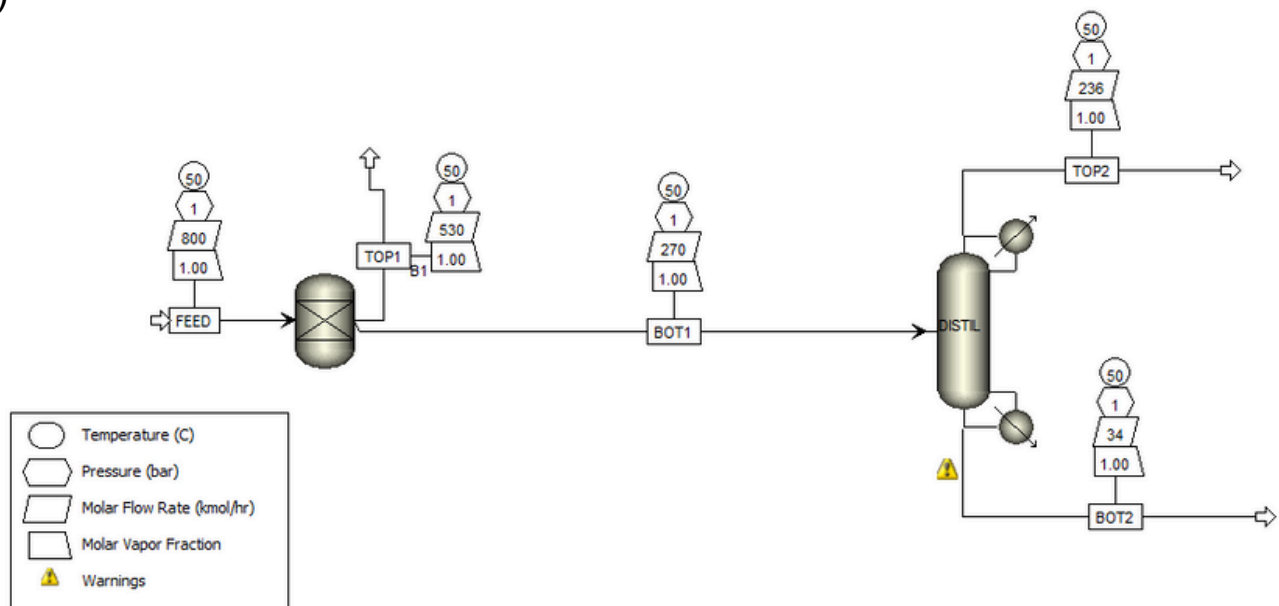
(b) indirect sequence (C₃/nC₄ separation first).



You must show the complete flowsheet for case (a) and case (b) scenarios including the process conditions (temperature, pressure and mass flow rates) of all streams embedded within the flow sheet with appropriate labelling (for reference see the illustration below).

Ans:

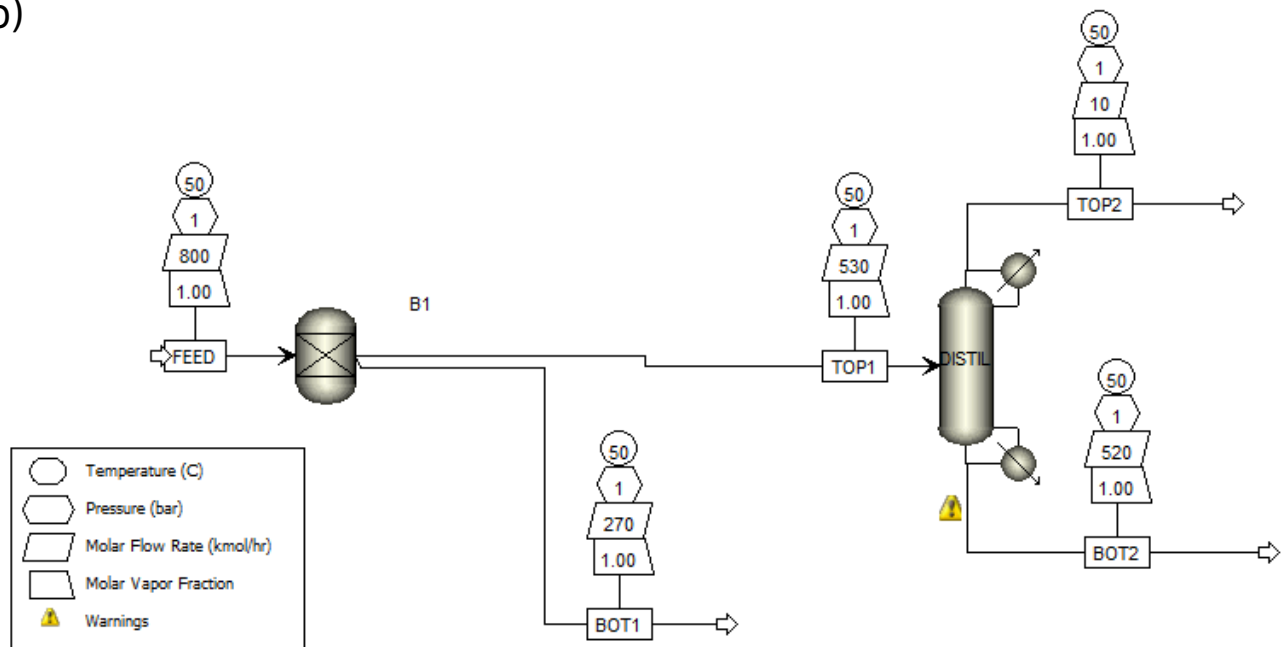
(a)



The Molar flows, mole fraction etc. are as following after running the simulation.

Stream Name	Units	Material				
		BOT1	BOT2	FEED	TOP1	TOP2
Mole Flows	kmol/hr	270	34	800	530	236
C1	kmol/hr	0	0	160	160	0
C2	kmol/hr	5	3.122502E-14	370	365	5
C3	kmol/hr	235	5	240	5	230
NC4	kmol/hr	25	24	25	0	1
NC5	kmol/hr	5	5	5	0	0
Mole Fractions						
C1		0	0	0.2	0.301886792	0
C2		0.018518519	9.18383E-16	0.4625	0.688679245	0.021186441
C3		0.87037037	0.147058824	0.3	0.009433962	0.974576271
NC4		0.092592593	0.705882353	0.03125	0	0.004237288
NC5		0.018518519	0.147058824	0.00625	0	0

(b)



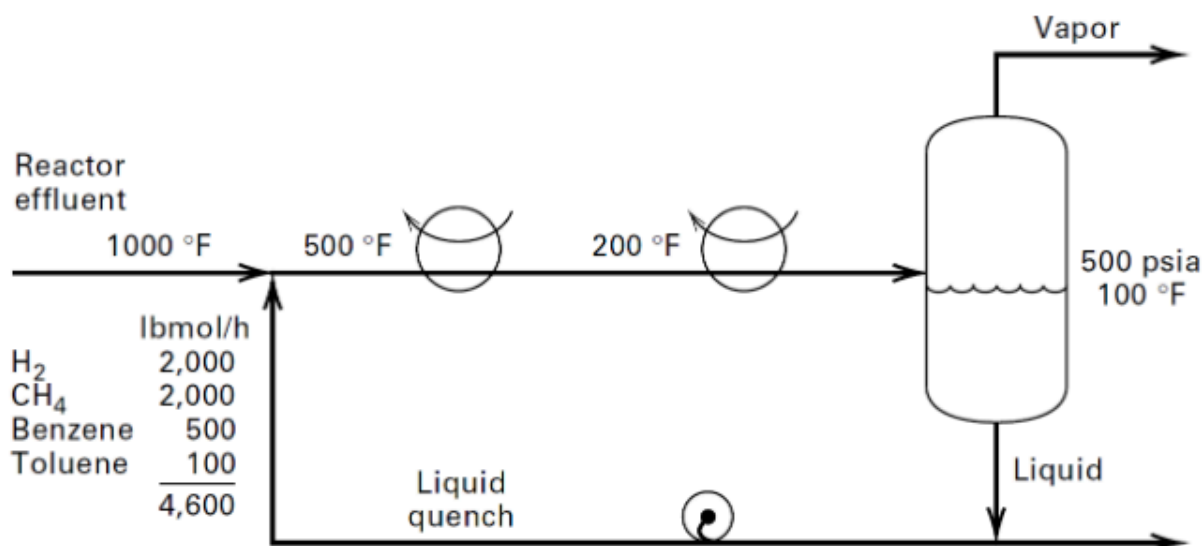
The Molar flows, mole fraction etc. are as following after running the simulation.

Stream Name	Units	Material			
		BOT2	FEED	TOP1	TOP2
Mole Flows	kmol/hr	520	800	530	10
C1	kmol/hr	160	160	160	0
C2	kmol/hr	360	370	365	5
C3	kmol/hr	0	240	5	5
NC4	kmol/hr	0	25	0	0
NC5	kmol/hr	0	5	0	0
Mole Fractions					
C1		0.307692308	0.2	0.301886792	0
C2		0.692307692	0.4625	0.688679245	0.5
C3		0	0.3	0.009433962	0.5
NC4		0	0.03125	0	0
NC5		0	0.00625	0	0

As we can see that for both cases the mole fractions are different which is expected as in first case we separated C2/C3 first and in the 2nd case we separated C3/nC4 first.

Q2: RECYCLE STREAM

In the following figure, a system to cool reactor effluent and separate light gases from hydrocarbons is shown. K-values at 500 psia and 100 °F are:

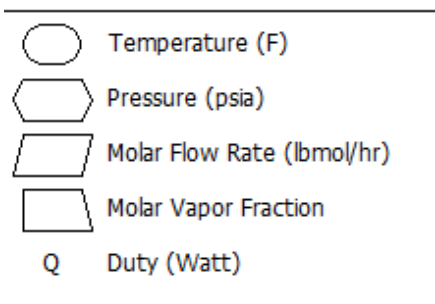
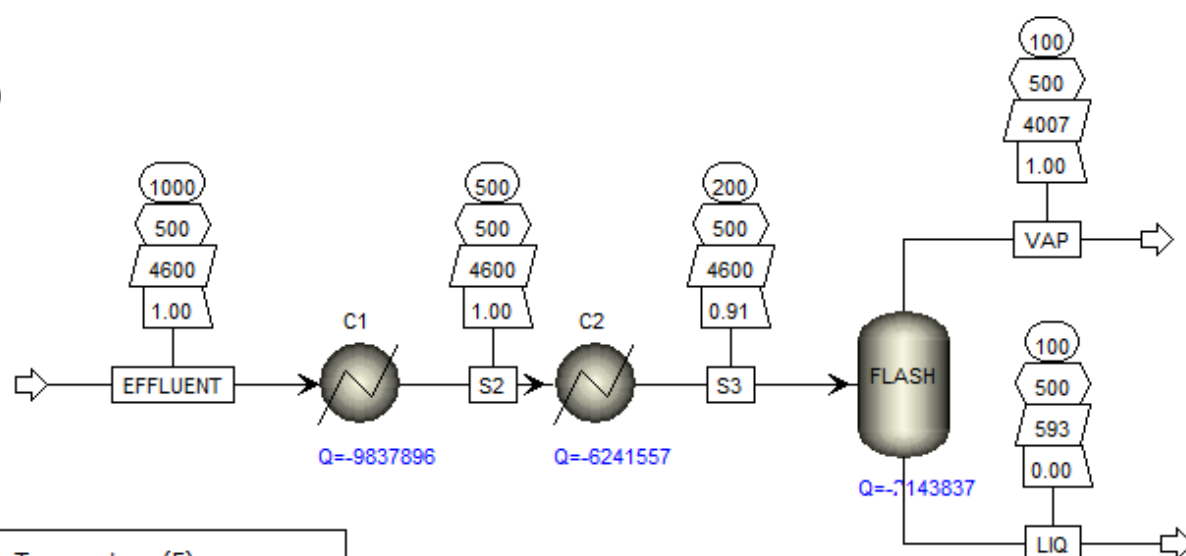


Component	K_i
H ₂	80
CH ₄	10
Benzene	0.010
Toluene	0.004

- Calculate composition and flow rate of vapor leaving the flash drum.
- Does the liquid-quench flow rate influence the result? Justify your answer.

Ans: This problem involves separating light gases (H₂, CH₄) from heavier hydrocarbons (benzene, toluene) using a flash drum. We need to find the vapor composition and flow rate leaving the drum using the given K-values. Additionally, we examine whether the liquid quench rate impacts the separation.

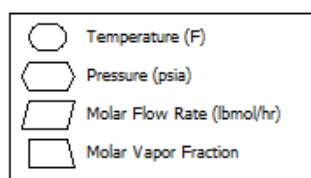
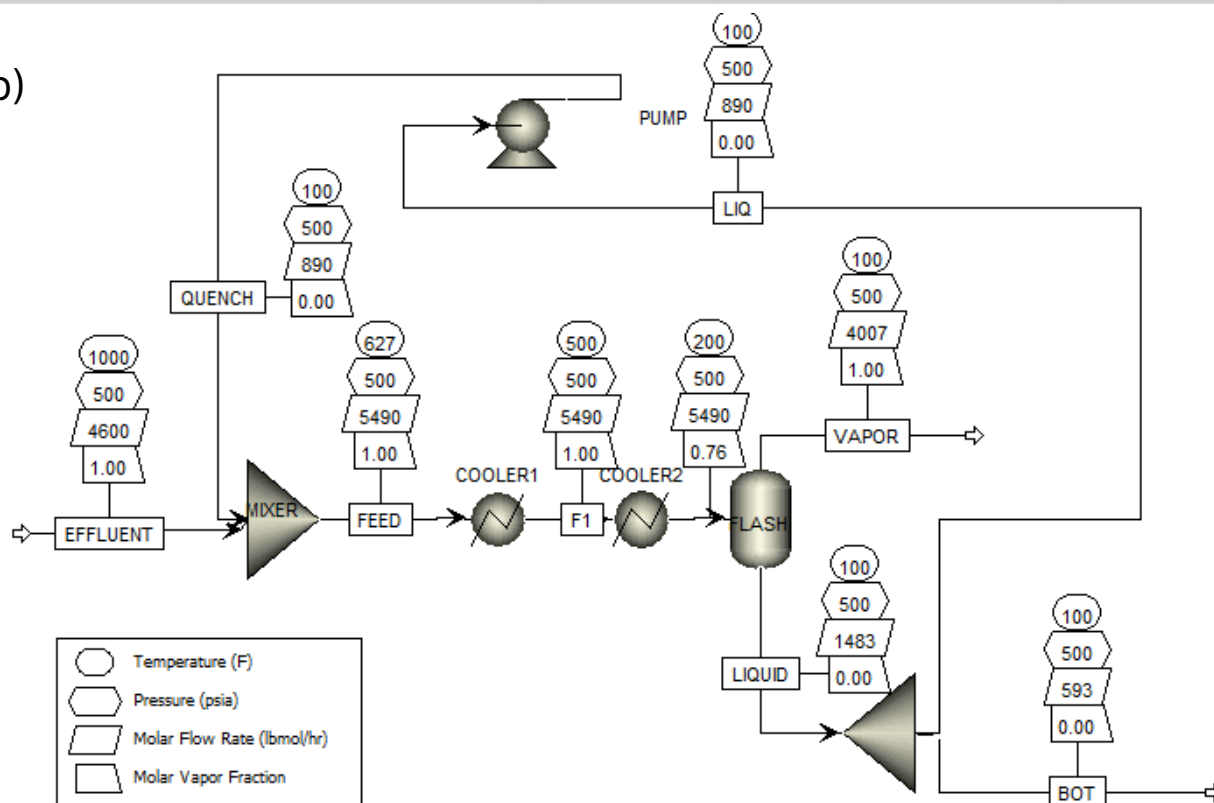
(a)



The Molar flows, mole fraction etc. are as following after running the simulation.

Material						
Stream Name	Units	EFFLUENT	LIQ	S2	S3	VAP
Mole Flows	lbmol/hr	4600	593.0311557	4600	4600	4006.968844
H2	lbmol/hr	2000	3.17204383	2000	2000	1996.827956
CH4	lbmol/hr	2000	23.7732913	2000	2000	1976.226709
BENZENE	lbmol/hr	500	468.3513606	500	500	31.64863936
TOLUENE	lbmol/hr	100	97.73445988	100	100	2.265540119
Mole Fractions						
H2		0.434782609	0.005348865	0.434782609	0.434782609	0.498338778
CH4		0.434782609	0.040087761	0.434782609	0.434782609	0.493197423
BENZENE		0.108695652	0.78975844	0.108695652	0.108695652	0.007898399
TOLUENE		0.02173913	0.164804933	0.02173913	0.02173913	0.0005654

(b)

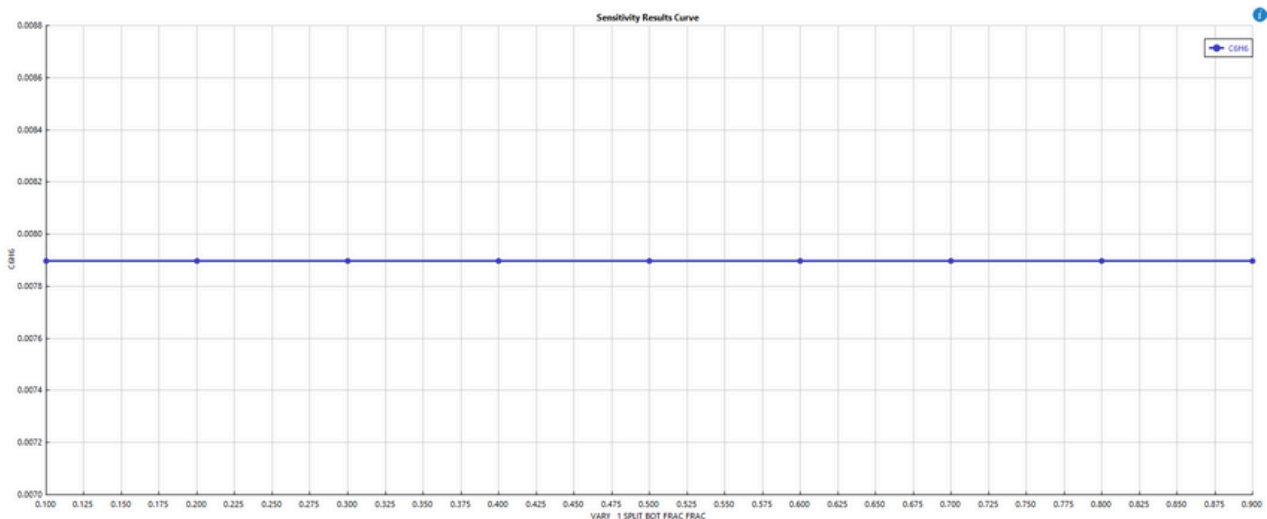


The Molar flows, mole fraction etc. are as following after running the simulation.

Stream Name	Units	Material				
		BOT	EFFLUENT	F1	F2	FEED
Mole Flows	lbmol/hr	593.0295516	4600		5489.544862	5489.544862
H2	lbmol/hr	3.172035326	2000		2004.757995	2004.757995
CH4	lbmol/hr	23.77324133	2000		2035.661985	2035.661985
BENZENE	lbmol/hr	468.3497718	500		1202.523076	1202.523076
TOLUENE	lbmol/hr	97.73450314	100		246.6018072	246.6018072
Mole Fractions						
H2		0.005348866	0.434782609		0.365195666	0.365195666
CH4		0.040087785	0.434782609		0.370825275	0.370825275
BENZENE		0.789757897	0.108695652		0.219056972	0.219056972
TOLUENE		0.164805452	0.02173913		0.044922086	0.044922086

LIQ	LIQUID	QUENCH	VAPOR
889.5443275	1482.573879	889.5443275	4006.970983
4.758052988	7.930088314	4.758052988	1996.827906
35.65986199	59.43310332	35.65986199	1976.228881
702.5246578	1170.87443	702.5246578	31.64864626
146.6017547	244.3362579	146.6017547	2.265549333
0.005348866	0.005348866	0.005348866	0.498338499
0.040087785	0.040087785	0.040087785	0.493197702
0.789757897	0.789757897	0.789757897	0.007898397
0.164805452	0.164805452	0.164805452	0.000565402

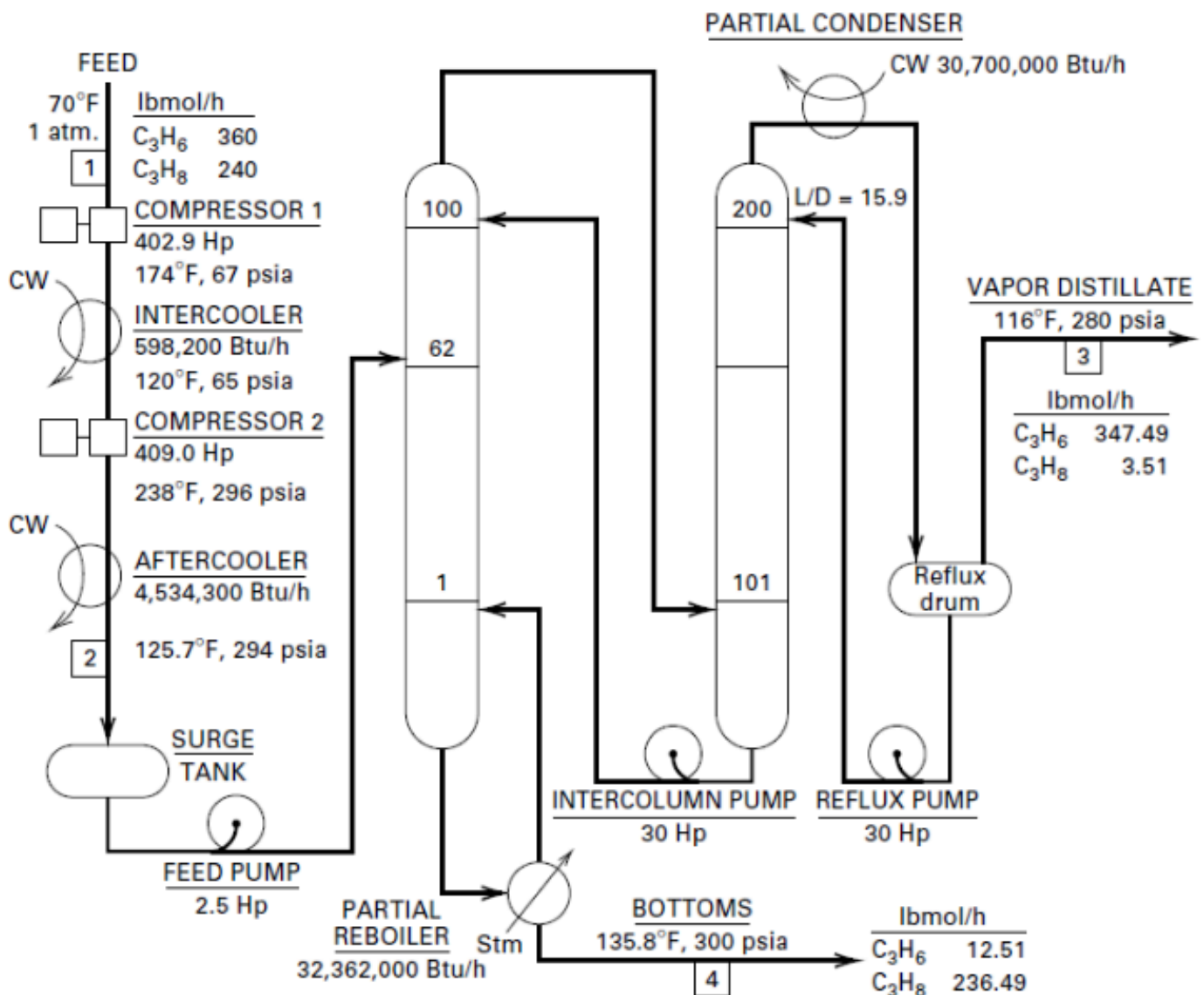
Further the sensitivity analysis was done to check whether there's significant impact of the quench stream or not.



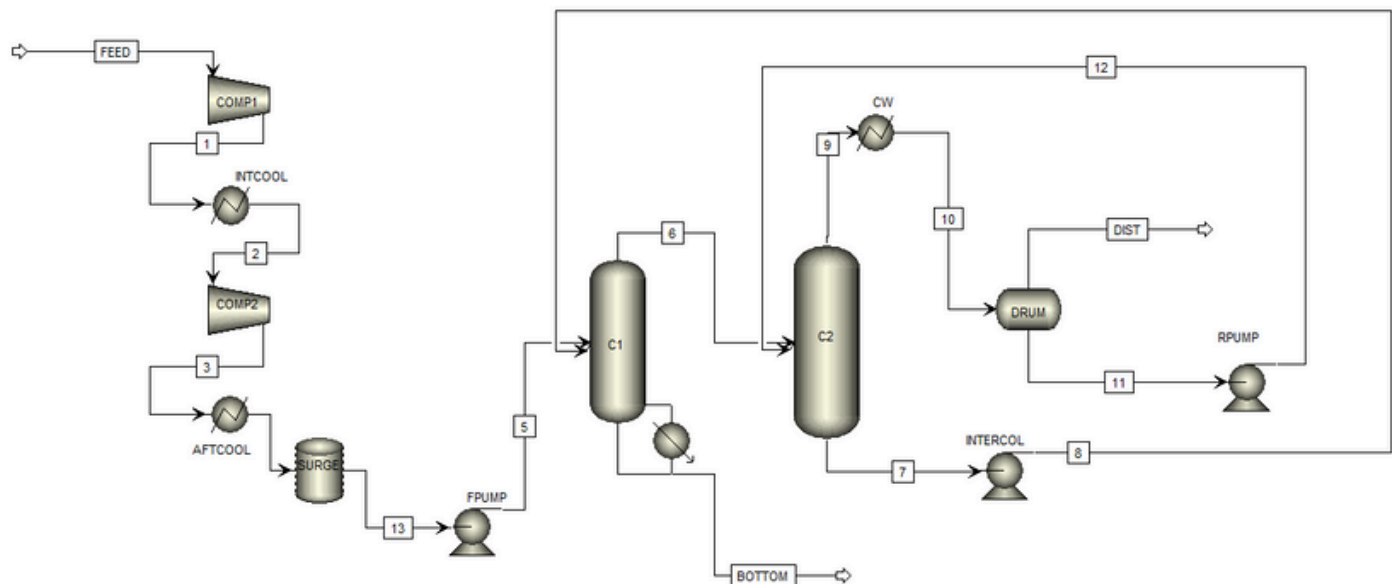
Finally we can see that quench stream did not change the benzene mole fraction significantly in the final output stream.

Q3: SEPARATION OF PROPYLENE FROM PROPANE

A process for the separation of a propylene-propane mixture to produce 99 mol% propylene and 95 mol% propane is shown. Because of the high product purities and the low α , 200 stages may be required. A tray efficiency of 100% and tray spacing of 24 inches will necessitate two columns in series, because a single tower would be too tall. Assume a vapor distillate pressure of 280 psia, a pressure drop of 0.1 psi per tray, and a 2-psi drop through the condenser. The stage numbers and reflux ratio shown are only approximate. Use AspenPlus with a real gas Equation of State to determine the necessary reflux ratio for the stage numbers shown in the respective columns.



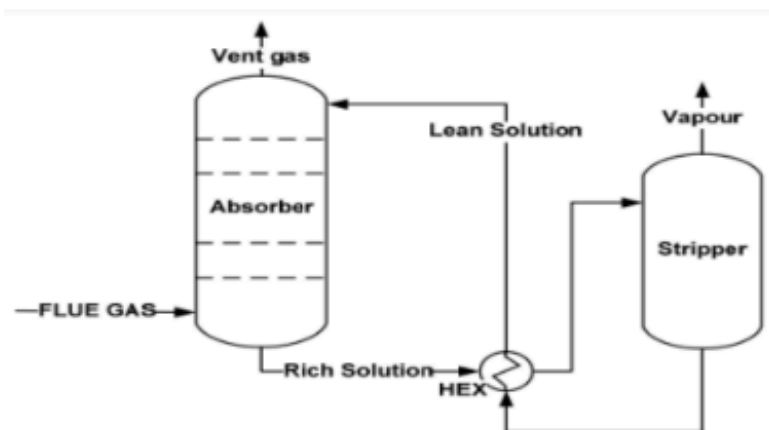
Ans: This problem focuses on the separation of propylene (C_3H_6) from propane (C_3H_8) using a two-column distillation system. The goal is to produce high-purity propylene (99 mol%) and propane (95 mol%) while managing the complexities of a large number of stages and high reflux ratios. Aspen Plus and a real gas Equation of State are recommended to determine the necessary reflux ratio for the given stage numbers. The problem highlights practical considerations like tray efficiency, pressure drops, and the limitations of tall column designs.



Q4: CO₂ CAPTURE FROM AIR (FLUE GAS)

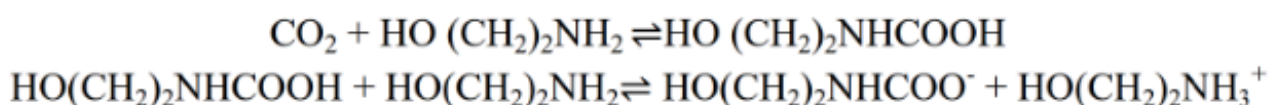
This is the basic operational flow diagram for amine absorption unit. The process is an exothermic, reversible reaction between a weak base (monoethanolamine, MEA) and a weak acid (CO₂) that leads to the formation of dissolved salts. In the absorber, the inlet gas is treated with a lean solvent. The solvent preferentially absorbs the acid gases. The CO₂-enriched solution is pre-heated before reaching the stripper, where the reaction is reversed by adding heat. The lean solvent escapes the stripper and is cooled by mixing heat with the rich solvent. The residual solvent is transferred to the absorber. A high purity CO₂ is extracted from the stripper's tip.

A huge amount of heat is mandatory for the regeneration of rich solvent.



Typically, the flue gas contains around 5 – 15% CO₂, by volume.

The overall forward reactions between CO₂ and MEA are-



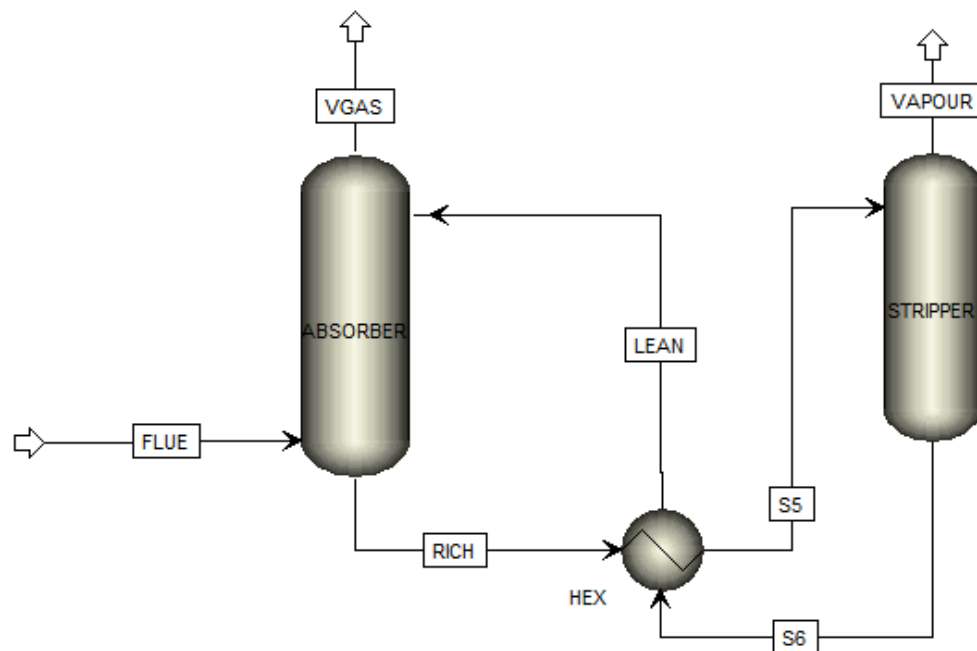
The first step is rate determining and second-order. The second phase is a reaction that occurs instantly.

Develop a AspenPlus simulation model, for

(a) Cutting-off the CO₂ emissions up to 85% by optimizing different parameters like MEA solvent, temperature, pressure, and reboiler duty.

(b) optimum reboiler duty by using heat of low pressure steam (waste steam) to reduce the operational cost.

Ans:



Q5: DESIGN OF A MEMBRANE ELECTROLYSER FOR GREEN H₂ PRODUCTION

Green hydrogen is produced by electrolyzing water, a process driven by renewable electricity sources like wind or solar power. The PEM electrolyser uses a solid polymer electrolyte to conduct protons from the anode to the cathode, where hydrogen gas is produced. This technology is favored for its rapid response times, high hydrogen purity, and compact design, making it suitable for integration with intermittent renewable energy sources. The Electrolyser simulation in Aspen Plus can be performed using the stack (applicable for part i and ii) component or the module (applicable for part iii).

(i) Perform (short-cut) simulation of the electrolyser and determine:

- Rate of production of H₂ and O₂
- Rate of water consumption and make-up water

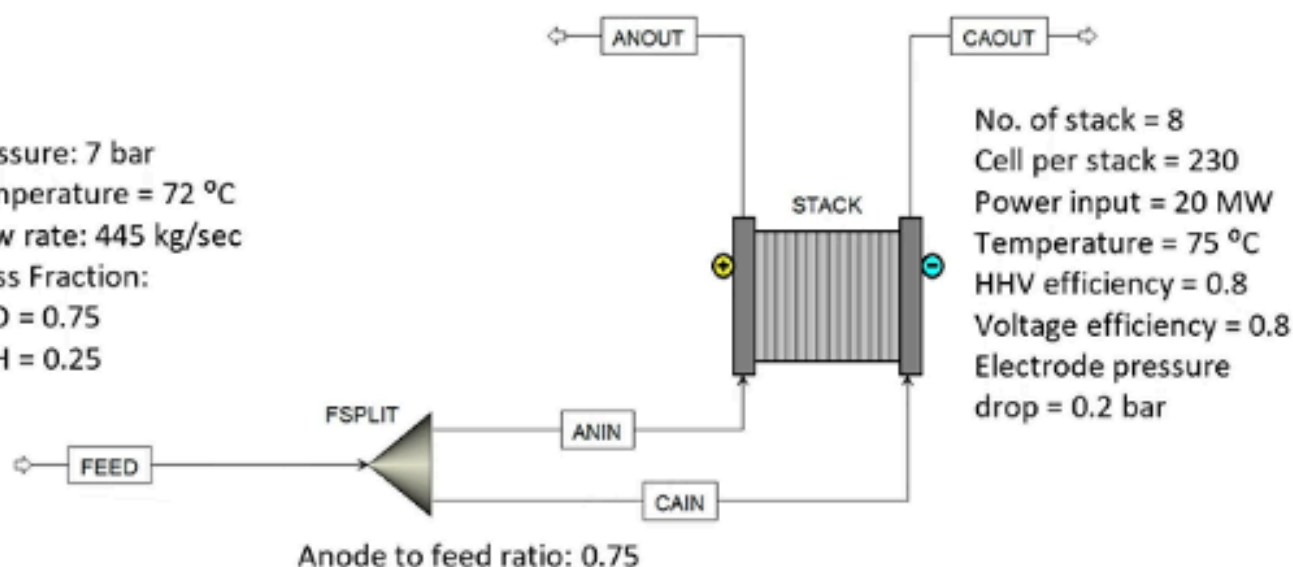
(ii) Perform rigorous simulation using the geometrical parameters given below and determine:

- Electrolyser efficiency
- Energy requirements per unit mass of H₂ production

(iii) Perform the same simulation using the electrolyser module and determine products.

- The makeup water flow rate (consider 96% purity of H₂ and O₂).

Pressure: 7 bar
 Temperature = 72 °C
 Flow rate: 445 kg/sec
 Mass Fraction:
 H₂O = 0.75
 KOH = 0.25



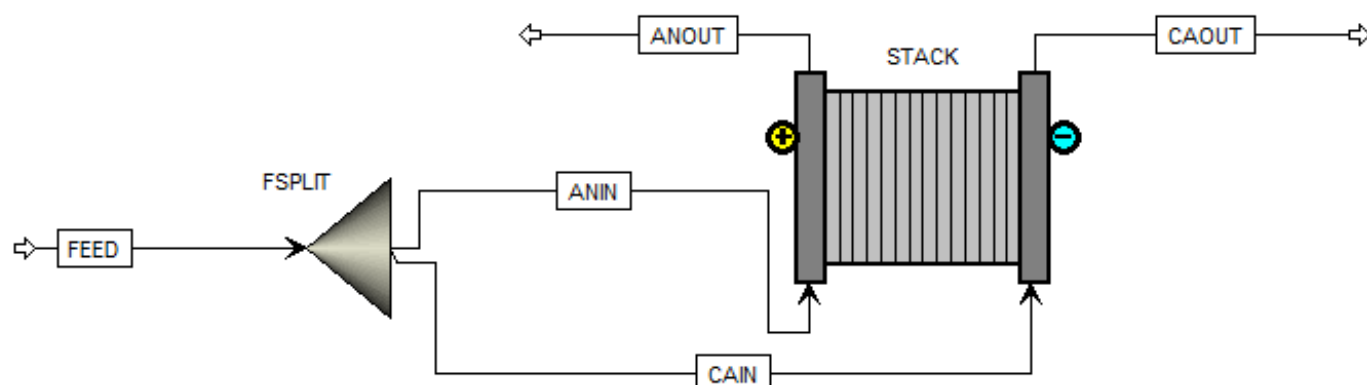
Electrode Parameters for Rigorous simulation:

Parameter	Anode	Units	Cathode	Units
Active area	4	sqm	4	sqm
Width of channel	2	meter	2	meter
Length of channel	2	meter	2	meter
Separation b/w electrode and separator	1.25	mm	1.25	mm
Thickness of channel	2	cm	2	cm
Thickness of electrode	2	mm	2	mm
Bubble zone width	0.5	mm	0.5	mm
Resistivity	7.00E-08	ohm-cm	7.00E-08	ohm-cm

Membrane Parameters

Parameter	Value	Units
Active area	4	sqm
Thickness (ts)	0.5	mm

Ans:
 (i)



Process and Model

Electrolysis: Alkaline water

Scope: Stack only

Reactions

Anode: $2OH^- = 0.5O_2 + H_2O + 2e^-$

Cathode: $2H_2O + 2e^- = H_2 + 2OH^-$

Overall: $H_2O = H_2 + 0.5O_2$

Operating Conditions

☒ Total power: 20 MW

☐ Total current: amp

Number of stacks: 8

Number of cells per stack: 230

☐ Total duty: Watt

☒ Temperature: 75 C

Pressure Drop Calculations

Specification type: ☒ Fixed value ☐ Correlation

Anode pressure drop: 0 bar

Cathode pressure drop: 0.2 bar

☒ Include liquid head in pressure drop calculation

The Molar flows, mole fraction etc. are as following after running the simulation.

Stream Name	Units	Material				
		ANIN	ANOUT	CAIN	CAOUT	FEED
Mole Flows	kmol/sec	15.38160511	15.46557462	5.127201702	5.071222029	20.50880681
H2O	kmol/sec	13.89445515	13.95043482	4.63148505	4.519525705	18.5259402
KOH	kmol/sec	1.487149955	1.487149955	0.495716652	0.495716652	1.982866607
H2	kmol/sec	0	0	0	0.055979673	0
O2	kmol/sec	0	0.027989836	0	0	0
Mole Fractions						
H2O		0.903316335	0.902031458	0.903316335	0.891210379	0.903316335
KOH		0.096683665	0.096158726	0.096683665	0.097750926	0.096683665
H2		0	0	0	0.011038695	0
O2		0	0.001809815	0	0	0

(ii)

Chemistry Specifications Equilibrium Constants Comments

Method of specifying chemistry

☒ Specify reactions ☐ Specify reactive components

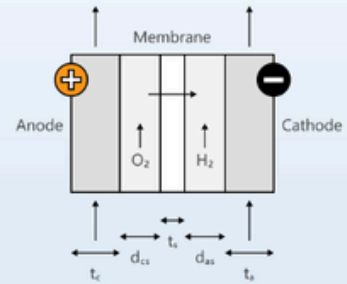
☐ Specify inert components ☐ All components are reactive

Reaction stoichiometry

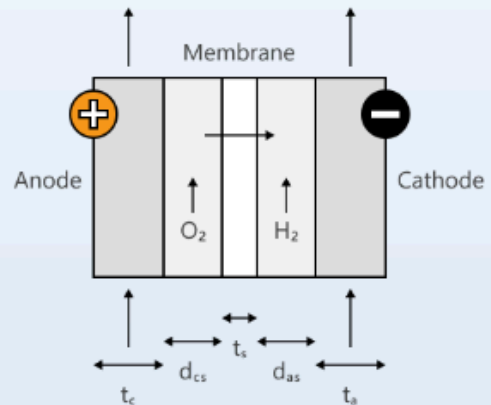
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	Reaction	Type	Stoichiometry	Delete
▶	1	Equilibrium	H2O <--> H2 + 0.5 O2	✗

Electrode Membrane Electrical Others					
Geometrical Parameters					
Parameter	Anode	Units	Cathode	Units	
Active area	4	sqm	4	sqm	
Width of channel	2	meter	2	meter	
Length of channel	2	meter	2	meter	
Porosity	0.3		0.3		
Tortuosity	3.8		3.8		
Pore radius	1e-06	meter	1e-06	meter	
Separation b/w electrode and separator (das, dcs)	1.25	mm	1.25	mm	
Thickness of channel (ta, tc)	2	cm	2	cm	
Electrode roughness factor	1.25		1.05		
Thickness of electrode	2	mm	2	mm	
Bubble zone width	0.5	mm	0.5	mm	
Channel roughness factor	1		1		
Activation free energy	8.051e+07	J/kmol	5.218e+07	J/kmol	



Membrane Parameters			
Parameter	Value	Units	
Active area	4	sqm	
Thickness (ts)	0.5	mm	
Porosity	0.42		
Tortuosity	2.18		
Wetness factor	0.85		
Oxygen diffusivity	1.81e-09	sqm/sec	
Hydrogen diffusivity	5.63e-09	sqm/sec	

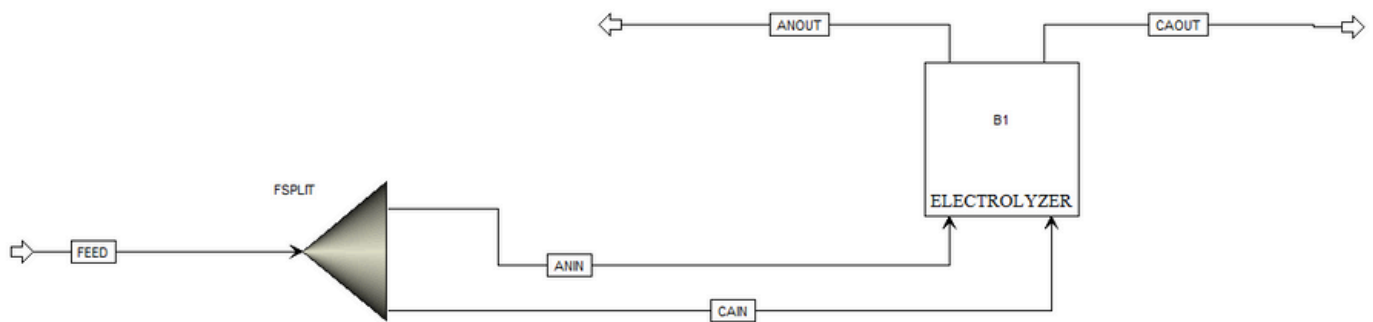


Material						
Stream Name	Units	ANIN	ANOUT	CAIN	CAOUT	FEED
Mole Flows	kmol/sec	15.38160511	15.81784427	5.127201702	4.720016334	20.50880681
H2O	kmol/sec	13.89445515	14.30157483	4.63148505	4.166256663	18.5259402
KOH	kmol/sec	1.487149955	1.48714981	0.495716652	0.495717647	1.982866607
H2	kmol/sec	0	6.581228E-05	0	0.058042025	0
O2	kmol/sec	0	0.029053823	0	4.901003E-11	0
Mole Fractions						
H2O		0.903316335	0.90414184	0.903316335	0.882678442	0.903316335
KOH		0.096683665	0.094017224	0.096683665	0.105024562	0.096683665
H2		0	4.160635E-06	0	0.012296997	0
O2		0	0.001836775	0	1.038344E-11	0

Efficiency: **0.7**

Energy requirement: **47.3 kWh/kg**

(iii)



Material						
Stream Name	Units	ANIN	ANOUT	CAIN	CAOUT	FEED
Mole Flows	kmol/sec	15.38160511	15.46557462	5.127201702	5.071222029	20.50880681
H2O	kmol/sec	13.89445515	13.95043482	4.63148505	4.519525705	18.5259402
KOH	kmol/sec	1.487149955	1.487149955	0.495716652	0.495716652	1.982866607
H2	kmol/sec	0	0	0	0.055979673	0
O2	kmol/sec	0	0.027989836	0	0	0
Mole Fractions						
H2O		0.903316335	0.902031458	0.903316335	0.891210379	0.903316335
KOH		0.096683665	0.096158726	0.096683665	0.097750926	0.096683665
H2		0	0	0	0.011038695	0
O2		0	0.001809815	0	0	0

Makeup water flowrate: **13.8946 Kmol/sec**